IMPERIAL COLLEGE LONDON

Faculty of Medicine

Faculty of Engineering

Department of Surgery and Cancer

Department of Bioengineering

TARGET-SPECIFIC BINDING AFFINITY PREDICTION-BASED SCORING FUNCTIONS FOR LEAD OPTIMIZATION

by

Chyi Ricketts
CID: 06071685

August 28th, 2026

under the supervision of
Dr. Pedro Ballester
Dr. Nivya James

IMPERIAL

Submitted in fulfillment of the Imperial College London MRes Cancer Informatics Degree Program

Target-Specific Binding Affinity Prediction-based Scoring Functions for Lead Optimization

Chyi Ricketts[1], **Nivya James**[2], **Pedro Ballester**[2]*
[1] Department of Surgery & Cancer, Faculty of Medicine, Hammersmith Hospital Campus, Imperial College London, London W12 0NN
[2] Department of Bioengineering, Faculty of Engineering, Imperial College London, White City Campus, London W12 0BZ

Abstract
Accurate prediction of protein–ligand binding affinity is a critical component of lead optimization in drug discovery, where small chemical modifications to a lead compound can result in meaningful changes in biological activity. While modern machine-learning scoring functions (SFs) perform well across diverse protein targets, they are often designed to prioritize generalization rather than target-specific performance. However, lead optimization is typically a target-focused problem involving congeneric series of compounds that differ by small structural modifications. This raises the question of whether target-specific models may provide improved performance compared to generalizable approaches, and whether the relevance of training data is as important as its quantity. This study investigates the tradeoff between dataset size and data relevance by developing and evaluating target-specific and family-specific binding affinity prediction (BAP) models for five cancer-related proteins: MCL1, SYK, HIF2α, PFKFB3, and MAPK14. Multiple machine-learning methods and featurizations were explored, with particular focus on ECIF::RF and AEV-PLIG::GATv2. Model performance was evaluated using the Ross FEP Benchmark, a dataset of protein targets and congeneric ligand series with experimental binding affinities. The results suggest an intermediate balance between retaining enough data to avoid overfitting and filtering enough data to preserve target-relevant information. This trend was most clearly observed with ECIF::RF, where four of five proteins showed peak performance at a structural similarity threshold of approximately 0.3–0.5. Target- and family-specific models also showed improved ranking and top-hit retrieval for several proteins, despite weaker absolute prediction compared with AEV-PLIG::GATv2. While these findings should be interpreted with caution given limited statistical significance and substantial target-to-target variability, they motivate further investigation into the role of data selection and relevance as an underexplored axis in developing efficient machine-learning models for lead optimization.

Background
The discovery of new drug candidates remains a resource-intensive process, with estimates suggesting that bringing a new therapeutic agent to market can require over a decade of development and investments exceeding 1 billion $ USD [1]. This challenge has continued to motivate computer-aided drug discovery (CADD) methods that supplement high-throughput screening (HTS), the identification and optimization of therapeutic compounds using screening assays [2, 3]. Among these methods, molecular docking and scoring functions are widely used to evaluate protein–ligand interactions. Molecular docking generates candidate binding conformations (poses) of a ligand within a protein binding site, while scoring functions estimate the strength of the resulting interaction [4]. This strength, or tightness of interaction between a potential drug and biomolecule, can be defined as binding affinity. Scoring functions (SFs) are commonly employed in structure-based virtual screening (SBVS) to prioritize active compounds from large chemical libraries, but are also used in later stages of drug discovery such as lead optimization, where binding affinity prediction (BAP) must be sufficiently sensitive to small chemical modifications in close analogs [5].

Lead optimization is an iterative stage of drug discovery in which promising compounds are refined to improve their therapeutic potential. Successful optimization requires balancing multiple properties, including binding affinity, potency, efficacy, and ADMET (Absorption, Distribution, Metabolism, Excretion, and Toxicity) characteristics. Among these, binding affinity is often, although not always, correlated with potency and specificity [6]. While improvements in binding affinity alone do not guarantee a successful drug candidate, accurate prediction of affinity changes can substantially accelerate the lead optimization process by enabling promising chemical modifications to be prioritized [6]. A variety of computational approaches have been developed to support lead optimization including traditional methods such as molecular docking [4], pharmacophore modeling [7], and free energy calculations [8, 9], while more recent machine-learning approaches seek to directly guide molecular design. For example, structure-based frameworks such as DeepFrag [10] and DELETE [11] utilize a masking strategy to propose chemical modifications such as fragment growing, linking, and replacement. Regardless of the strategy used to generate new compounds, accurate estimation of the resulting changes in binding affinity remains a central and extremely challenging task in lead optimization [12].

For many years, classical scoring functions dominated the field as a means of rapidly estimating binding affinity during molecular docking and can generally be categorized as force-field-based [13], empirical [14], and knowledge-based approaches [15, 16]. Force-field methods estimate intermolecular interactions using physically motivated energy terms, empirical methods fit weighted interaction terms to experimental affinity data, and knowledge-based methods derive interaction potentials from statistical analyses of experimentally determined protein–ligand structures. However, biological binding processes are often too complex to be accurately captured by the predetermined functional forms assumed by classical scoring functions [17]. To improve physical realism and predictive accuracy, physics-based free-energy methods were subsequently developed. These methods span a spectrum of increasing rigor, ranging from docking-based energy evaluation and endpoint free-energy methods to fully alchemical binding free-energy (BFE) calculations. Among these, alchemical approaches such as Free Energy Perturbation (FEP) and Thermodynamic Integration (TI) are widely regarded as some of the most accurate computational techniques currently available for predicting relative binding affinities [18, 19]. FEP is particularly well suited to lead optimization applications, where compounds are often organized into congeneric series differing by small chemical modifications. However, despite their accuracy, physics-based methods remain computationally demanding, which can substantially limit their throughput and practical application during iterative lead optimization. In particular, FEP+ has a high computational cost relative to ML SFs, with AEV-PLIG reported to be approximately 400,000 times faster than FEP+ [20]. Physics-based methods are also sensitive to factors such as ligand parameterization, force-field selection, and sampling convergence [21]. Together, these limitations have motivated increasing interest in machine-learning (ML) approaches as a means of achieving comparable predictive performance at substantially lower computational cost. ML scoring functions differ fundamentally from classical approaches because they do not rely on predefined functional forms. Instead, they can implicitly capture intermolecular binding interactions that are difficult to model explicitly [17], and have been shown to outperform a broad range of classical scoring functions on established benchmarking datasets [12].

Currently, many scoring functions prioritize generalization, which is essential for structure-based virtual screening across diverse targets and chemical spaces. This broad applicability is advantageous in early-stage discovery, where target diversity and limited prior information make general models more robust and data-efficient. Among these approaches, AEV-PLIG represents one of the most advanced structure-based scoring frameworks. It combines radial Atomic Environment Vector (AEV) descriptors with a graph neural network (GNN) architecture, the Protein–Ligand Interaction Graph (PLIG), to model detailed geometric and chemical environments of protein–ligand complexes. By training on large and structurally diverse datasets, AEV-PLIG is designed to maximize generalization performance across targets and chemical spaces [20].

However, in lead optimization, the problem is fundamentally different: compounds are typically confined to a narrow, congeneric chemical series around a known scaffold, and models are repeatedly applied to predict small, fine-grained changes in binding affinity for a single target [22, 23]. Another particularly challenging aspect within congeneric series is the presence of activity cliffs, pairs of structurally similar compounds that exhibit large differences in binding affinity [24, 25]. Such cases are difficult for generalizable models to capture, as the relationship between structural and affinity similarity breaks down, potentially favoring target-specific approaches trained on relevant chemical series. More generally, this reflects the broader trade-off between data relevance and data quantity. While large and diverse datasets generally improve robustness and reduce variance, they may also dilute target-specific signals that are critical for accurate relative binding affinity prediction. Conversely, training on smaller but highly relevant datasets can improve sensitivity to local structure activity relationship (SAR) trends, but increases the risk of overfitting and poor generalization. This tension has motivated approaches such as transfer learning, multi-task learning, and active learning, which aim to combine global chemical knowledge with local target adaptation [26, 27]. In this context, determining whether "more relevant data" can outperform "more data" depends on balancing representational breadth with chemical and biological specificity. Understanding this trade-off is particularly important for lead optimization, where predictive accuracy within a narrow chemical series is often more valuable than broad applicability across unrelated targets.

One of the most recent challenges in machine-learning-based scoring functions is fair and unbiased evaluation [28]. Many commonly used benchmark datasets exhibit overlap in protein targets, ligand scaffolds, or chemical series between training and test sets, which can lead to inflated estimates of predictive performance [29]. To address these limitations, this study employs the Ross FEP Benchmark (2023), a large dataset of protein–ligand congeneric series with experimentally measured binding affinities and corresponding FEP+ predictions [18]. Because lead optimization typically involves predicting relative affinity changes within congeneric series, this benchmark provides a particularly relevant and challenging evaluation setting for binding affinity prediction models.

This study focuses on target-specific binding affinity prediction using machine-learning scoring functions, where improved target-level accuracy may benefit lead optimization. To investigate the trade-off between dataset size and data relevance, models are trained using varying amounts of data across multiple cancer-related targets: MCL1, SYK, HIF2α, PFKFB3, and MAPK14. The selection of these five proteins were guided by two clear criteria. First, all give are cancer-relevant targets with known inhibitors and varying degrees of clinical translation, ranging from approved therapeutics to compounds in active clinical trials. Second, each contributes a congeneric series of sufficient size to the Ross FEP Benchmark. Together, these five proteins also span a range of structural and data-richness contexts, from well-characterized systems such as MCL1 to more challenging and less-annotated targets such as PFKFB3, providing a diverse basis to evaluate different modeling approaches respond to changes in dataset size and feature representation.

Material and Methods

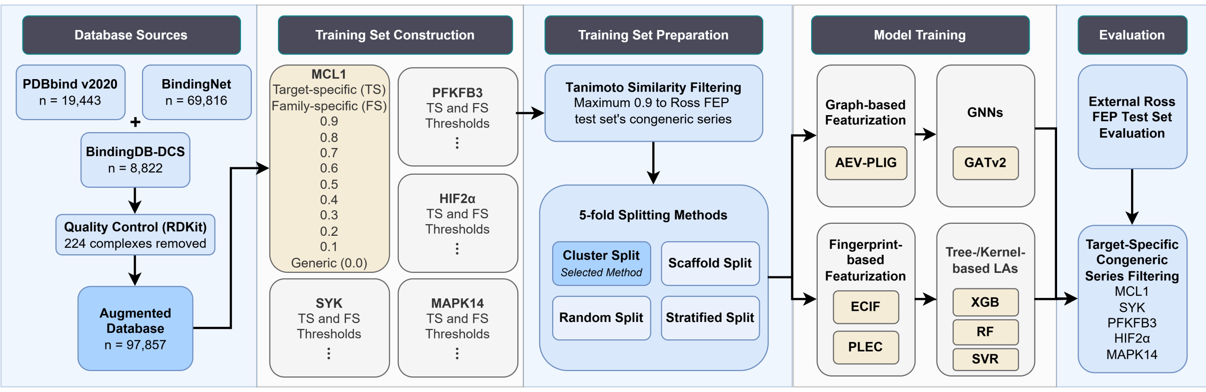


Figure 1: Methods Flowchart. Overview of the complete modeling pipeline from data sources to evaluation. Each model uses the augmented database unless otherwise specified. All five proteins include both a Target-specific (TS) training set and Family-specific (FS) training sets at thresholds from 0.1 to 0.9, with 0.0 (Generic) as the baseline; MCL1 is shown in full as a representative example. Each model contains a featurization scheme and supervised learning algorithm, and is hereafter referred to in the format Featurization::Learning Algorithm (e.g. ECIF::XGB). Each model is evaluated on the respective congeneric series in the Ross FEP Test Set.

Training Databases

Database selection is crucial in structure-based binding affinity prediction, as the quality, chemical diversity, and distribution of protein–ligand complexes directly influence learned feature representations and model generalization. Accordingly, three widely used datasets are employed for training and validation: PDBbind v2020 [30], BindingNet [31], and BindingDB-DCS [32]. Binding affinity measurements (K_i and K_d) were converted to binding free energies (ΔG, kcal/mol) using $\Delta G = -1.36 \log_{10}(K)$ at $T = 298$ K. Database sizes are notated in Figure 1 and additional details on database construction and preprocessing are provided in the Supplementary Methods.

Ross FEP Test Set

The Ross FEP Benchmark [18] serves as the external test set for all models in this study. The dataset contains 1,184 complexes across 92 congeneric series, each with experimentally measured binding free energies (ΔG) and corresponding predictions from the physics-based alchemical relative binding affinity method FEP+. Each protein of interest (MCL1, SYK, HIF2α, PFKFB3, and MAPK14) contributes one congeneric series to this test set. Since lead optimization typically involves predicting relative affinity changes within congeneric series, this benchmark provides a particularly relevant and challenging evaluation setting for binding affinity prediction models. Crucially, the Ross FEP Test Set is held out entirely from all training and validation procedures; no splitting strategy is applied to the test data.

Tanimoto Similarity Filtering

To reduce model memorization, all training set complexes with a Tanimoto similarity greater than 0.9 to any ligand in the target's congeneric series in the Ross FEP Test Set are removed. Similarity was computed using 2048-bit ECFP6 fingerprints [33].

Dataset Splitting

Four 5-fold splitting techniques were evaluated: random split, stratified split, scaffold split, and cluster split. The 5-fold cluster split was selected as it both maintained representativeness of the test set while minimizing ligand similarity between training and validation sets (Supplementary Figure 1). Cluster split is performed with the Butina algorithm [34] a greedy, centroid-based method that groups compounds into highly similar clusters using a Tanimoto cutoff of 0.8 to the cluster centroid based on pairwise structural similarity. Scaffold splitting was based on Bemis-Murcko scaffolds [35], which enforce structural separation of the core chemical framework across splits. Consequently, all subsequent model complexity experiments were performed using a 5-fold Butina cluster split with a threshold of 0.8. As the Ross FEP benchmark is an external test set, no splitting procedure was applied. All split strategy exploration is evaluated by validation set metrics as to reduce the risk of biased selection and overfitting [28].

TM-align

TM-align is a widely-used algorithm for fast and accurate protein structure alignment that optimizes both residue-residue and topology-based structural similarity [36]. It reports TM-scores ranging from 0 to 1 to quantify topological similarity between protein structures. TM-align processing excluded four receptor structures (3v3q, 5a3n, 5mwp, and 5mwy) due to receptor sequences shorter than 3 residues, representing 40 complexes total from BindingNet (36 from 5wwp alone). All remaining complexes successfully completed structural alignment.

Training Set Construction

For each protein of interest, two types of training sets were constructed. Target-specific (TS) datasets contain only complexes whose receptor structure exactly matches the target protein, identified by UniProt ID / PDB ID matching. Family-specific (FS) datasets were constructed using TM-align structural similarity scores between each training complex receptor and the target receptor found in the corresponding congeneric series. A training complex was included in the family-specific dataset at threshold θ if its TM-score $\geq \theta$, where $\theta \in \{0.0, 0.1, ..., 0.9\}$. Generic models are equivalent to those with a TM-score of 0.0 and therefore retain all complexes regardless of structural similarity. It is important to note that target-specific training sets are not necessarily a strict subset of high-threshold family-specific datasets: the reference receptor structures used in the Ross FEP Test Set may represent only a partial domain of the full protein sequence, resulting in imperfect TM-score matches to full-length structures in the training database. Consequently, some training complexes that are biologically identical to the target may receive TM-scores below a given threshold θ and be excluded from the corresponding family-specific dataset, while remaining present in the target-specific set.

Featurization Schemes

AEV-PLIG is a GNN-based scoring function that combines radial Atomic Environment Vector (AEV) descriptors with a Graph Attention Network (GATv2) architecture to model protein–ligand interactions [20]. This study compares this graph-based representation to two fingerprint-based methods: ECIF and PLEC. ECIF (Extended Connectivity Interaction Fingerprints) [37] encodes protein–ligand interactions as atom-pair counts within a defined distance cutoff, capturing the chemical environment of the binding site using a distance cutoff of 6.0 Å. PLEC (Protein–Ligand Extended Connectivity) fingerprints [38] encode protein-ligand contact environments using extended connectivity fingerprints applied to the interaction interface, computed with protein depth = 4, ligand depth = 2, and a distance cutoff of 4.5 Å.

Learning Algorithms

GATv2 [39] is a dynamic graph attention network that computes attention coefficients dependent on both the source and target node features, enabling more expressive and flexible aggregation of neighborhood information compared to the original GAT architecture. AEV-PLIG is used with a GATv2 learning algorithm and is hereafter referred to as AEV-PLIG::GATv2. GATv2 architecture choices are reported in Supplementary Figure 2. Three regression/kernel-based models are used: Random Forest (RF) [40], Extreme Gradient Boosting (XGB) [41], and Support Vector Regression (SVR) [42]. Random Forest is an ensemble of decision trees trained on a random bootstrapped sample of the data, aggregated on the mean to give robust predictions. XGB is a gradient boosting framework that sequentially builds K decision trees, each correcting the residual errors of the previous ensemble. SVR finds a function that fits the training data within a specified ε-insensitive margin, using an RBF kernel to model non-linear relationships.

Hyperparameter Tuning

Optuna is an open-source hyperparameter optimization framework [43] that utilizes Bayesian-style samplers, namely Tree-structured Parzen Estimator (TPE), for efficient selection from a specified search space over a finite number of trials. TPE uses kernel density estimation (Parzen estimators) to model the distributions of hyperparameters associated with successful versus unsuccessful trials, and subsequently proposes new configurations by maximizing an acquisition function that balances exploration and exploitation. Each tree-based or regression model within this study was tuned using 30 trials within a search space specified in Supplementary Table 3. All hyperparameter tuning is evaluated by validation set metrics as to reduce the risk of biased selection and overfitting [28]. All selected hyperparameters are available at `https://github.com/chyiricketts/Target-Specific_BAP_Prediction`.

Performance Metrics

Model performance is evaluated using four complementary metrics: Pearson Correlation Coefficient (PCC), Kendall's Tau (Kτ), Root mean squared errors (RMSE), and Overlap@K. Full definitions and equations are provided in Supplementary Methods.

Statistical Testing

To assess model performance uncertainty, a 1,000-iteration bootstrap procedure was employed to resample compounds with replacement from the corresponding congeneric series. The 95% bootstrap confidence intervals are reported as error bars in relevant figures. Secondly, to assess whether differences in performance between models were statistically significant, two sets of pairwise one-sided 1,000-iteration bootstrap comparisons were conducted. The first compared each model against the Generic AEV-PLIG::GATv2 reference; the second compared each family-specific or target-specific model against its own family's generic (threshold = 0.0) configuration, isolating the effect of relevance filtering independent of architecture. The p-value reported is the fraction of bootstrap samples in which the reference model scored higher. Full definitions and equations are provided in the Supplementary Methods while Kτ p-values are reported in full for ECIF::RF and ECIF::XGB in Supplementary Tables 4 and 5.

Results

Target-focused Dataset Curation

For each protein of interest, target-focused training sets were constructed consisting of (i) target-specific datasets containing exact protein matches and (ii) family-specific datasets derived using TM-align structural similarity thresholds. Figure 2A shows the size of each target-specific dataset alongside the corresponding congeneric series in the Ross FEP Test Set. Figure 2B shows the structural similarity distributions for each protein while exact family-specific dataset sizes are reported in Supplementary Table 1. This representation allows us to see that MAPK14 and SYK exhibit high structural conservation (>10,000 complexes at TM-score >0.6), reflecting conserved kinase folds, whereas PFKFB3 and HIF2α show markedly lower similarity to other training database proteins, with PFKFB3 retaining fewer than 15 complexes above a threshold of 0.4, likely reflecting their distinct overall folds and limited representation in publicly available databases.

A

Dataset	Original Size	MCL1	SYK	PFKFB3	HIF2α	MAPK14
PDBbind	19,308	41	50	4	14	149
BindingNet	69,727	264	550	70	55	1,562
BindingDB	8,822	0	3	0	0	1
Augmented	**97,857**	**305**	**603**	**74**	**69**	**1,712**
Ross FEP Test Set	1,184	42	46	40	41	34

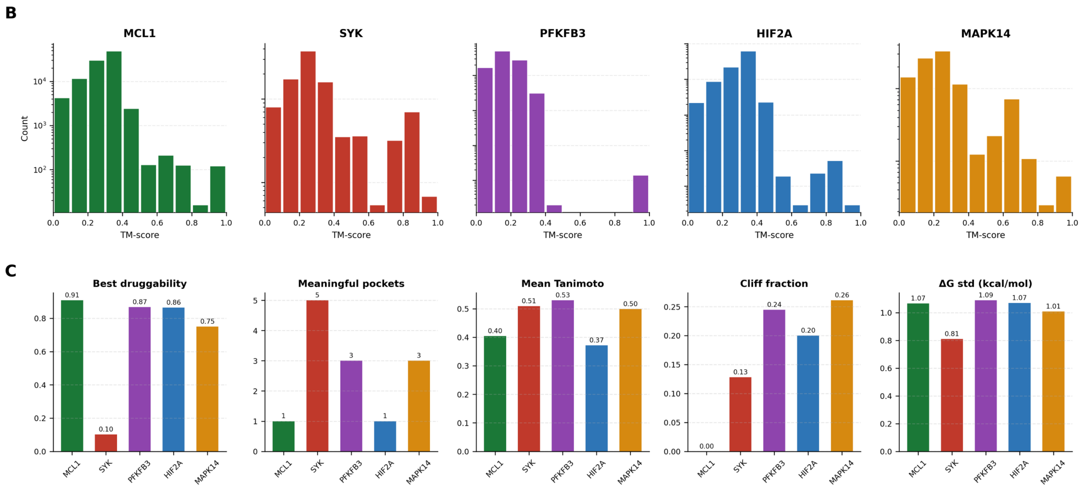


Figure 2: Training and Test Set Sizes and Molecular Property Distributions. (a) Number of protein–ligand complexes contributed by each source database (PDBbind, BindingNet, BindingDB-DCS) to the target-specific training sets and aggregated Augmented database, alongside the size of each protein's congeneric series in the Ross FEP Test Set. (b) TM-score distributions (log scale) of structurally similar complexes in the Augmented database relative to each protein of interest, illustrating the varying availability of structurally relevant training data across targets. MAPK14 and SYK show abundant high-similarity complexes reflecting conserved kinase folds, while PFKFB3 and HIF2α drop sharply at higher thresholds. (c) Binding-site and chemical series characteristics of each test set protein: best druggability score (fpocket), number of meaningful pockets (meaningful defined as fpocket score ≥ 0.1), mean pairwise Tanimoto similarity, activity cliff fraction, and ΔG standard deviation (kcal/mol).

Figure 2C summarizes binding-site and chemical series characteristics of each test set protein [44]. Each pocket receives a druggability score from 0 to 1, computed via a logistic function over selected pocket descriptors, where values above 0.5 suggest plausible drug-like binding Pockets with a druggability score of ≥ 0.1 are considered meaningful in this study. A greater number of meaningful pockets indicates a more ambiguous binding site, as the model must implicitly learn which pocket is relevant without explicit guidance, potentially complicating affinity prediction. Here, MCL1 presents a single, well-defined binding pocket (druggability score 0.91 over one meaningful pocket) and low activity cliffs, suggesting a relatively tractable prediction target. In contrast, SYK has a low best druggability score (0.10) and five meaningful pockets, indicating a more ambiguous binding site that may complicate affinity prediction. SYK's low druggability score likely reflects its apo crystal structure, in which the binding site appears geometrically fragmented rather than presenting as a single well-defined cavity, rather than true biological undruggability given the existence of approved SYK inhibitors such as fostamatinib (Additional information on protein inhibitors and clinical stage in Supplementary Table 2). PFKFB3 and MAPK14 exhibit the highest cliff fractions (0.24 and 0.26 respectively), indicating that structurally similar compounds can differ substantially in binding affinity which is known to be difficult for ML BAP. Mean pairwise Tanimoto similarities across congeneric series range from 0.37 to 0.53, consistent with the structurally focused but chemically diverse nature of congeneric series used in lead optimization. Experimental ΔG standard deviations are comparable across proteins (0.81–1.09 kcal/mol), indicating similar dynamic ranges in binding affinity.

Figure 3: Target-specific Model Performance. (A) PCC, (B) Kτ (C) RMSE and (D) Overlap@5 for each protein (y-axis) across all target-specific models, including ECIF and PLEC fingerprint featurizations with learning algorithms XGB, RF, and SVR. Generic AEV-PLIG::GATv2 and Target-specific AEV-PLIG::GATv2 performance are shown for comparison. Values are point estimates from the full congeneric test series, without bootstrap resampling, averaged across seed/fold replicates.

Target-specific Model Performance

Figure 3 shows a full comparison of all target-specific models across all four performance metrics. Across all four metrics, no single model consistently dominated, though clear trends emerged. In PCC (Figure 3A), Generic AEV-PLIG::GATv2 achieves the highest performance for the five proteins, reflecting its stronger absolute prediction accuracy when trained on the full augmented dataset.

In Kendall's τ (Figure 3B), ECIF::RF led for three of five proteins, suggesting stronger ranking performance despite weaker absolute predictions. RMSE (Figure 3C) results were mixed, with Generic AEV-PLIG::GATv2 generally outperforming the rest of the model while ECIF::RF, ECIF::XGB, and ECIF::SVR remained competitive. In Overlap@5 (Figure 3D), generic AEV-PLIG::GATv2's weakness emerges and only wins out on SYK. For the remaining proteins, ECIF::RF dominated across four proteins. The remaining fingerprint::regression/kernel-based models showed reasonable Overlap@5 on all proteins with the exception of MAPK14. Across all five targets, ECIF-based models generally outperformed their PLEC counterparts.

The poor performance of target-specific AEV-PLIG::GATv2 relative to its generic counterpart suggests that high-capacity GNN architectures may not be well-suited to this data regime, where target-specific training sets are substantially smaller. This motivates further exploration of lower-complexity and more regularized approaches.

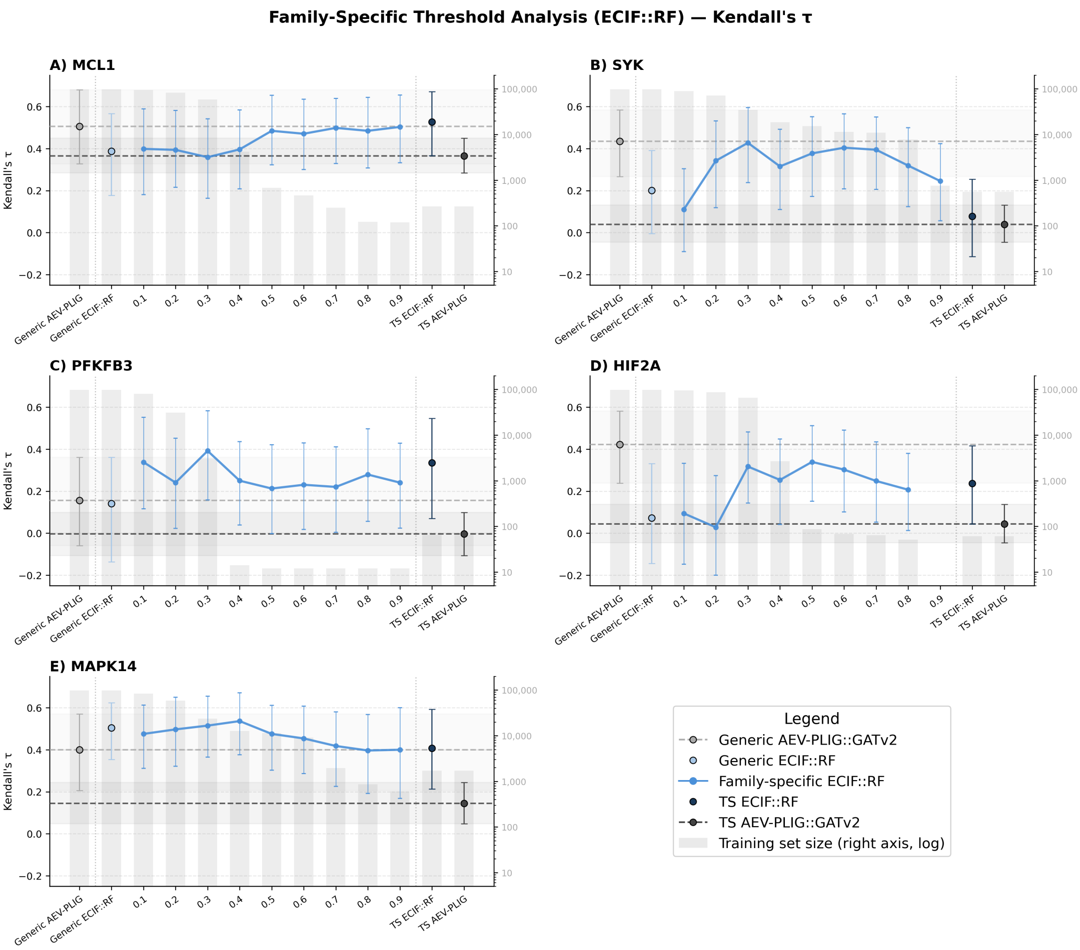


Figure 4: Family-specific Model Performance for ECIF::RF. For each protein, (a) MCL1, (b) SYK, (c) PFKFB3, (d) HIF2α, and (e) MAPK14, ECIF::RF models were trained on family-specific training sets at structural similarity thresholds from 0.1 to 0.9, alongside a generic model (threshold = 0.0) as the baseline. Comparisons are shown for Generic and Target-specific AEV-PLIG::GATv2 and Generic and Target-specific ECIF::RF models. Gray bars (right-side log scale) indicate training set size at each threshold, decreasing as the structural similarity threshold becomes more stringent. Note that target-specific training sets are not necessarily smaller than the 0.9 family-specific dataset (see Training Set Construction). The primary metric shown is Kendall's τ; additional metrics (PCC, RMSE, Overlap@5) are provided in Supplementary Figures 6.All point estimates represent metrics computed once on the full congeneric test series using the canonical ensemble prediction. Uncertainty is quantified as 95% bootstrap confidence intervals from 1,000 iterations of resampling with replacement, computed for all models from the same procedure. No pairwise comparisons reached statistical significance ($p < 0.05$). HIF2α is missing the family-specific threshold of 0.9 due to insufficient training set size.

Model performance varied substantially across protein targets. MCL1 consistently achieves the highest predictive performance across metrics, suggesting that its binding affinity relationships are more readily captured by the investigated models, potentially reflecting its strong representation in the training data and well-defined binding site (Figure 2C). In contrast, SYK generally exhibits lower predictive performance despite its large target-specific training set, indicating a more challenging prediction task potentially attributable to its multiple meaningful pockets and ambiguous binding site (Figure 2C). PFKFB3 produces comparatively weak results across most model configurations, consistent with its limited training data availability and distinct protein fold.

Family-specific Training Sets

Figure 4 shows the generic, target-specific, and all family-specific thresholds for ECIF::RF across all five proteins, with Generic and Target-specific AEV-PLIG::GATv2 shown as reference.

Consistent with the target-specific results (Figure 3), target-specific ECIF::RF outperforms target-specific AEV-PLIG::GATv2 across all proteins, reflecting the greater susceptibility of high-capacity GNN architectures to overfitting when training data are limited. This robustness is reflected in the family-specific training regime. As the structural similarity threshold increases and training set size decreases, ECIF::RF performance remains comparatively stable, whereas AEV-PLIG::GATv2 variance increases substantially (Supplementary Figure 8), from a standard deviation of 0.054 for the generic model to 0.161 for the 0.9 FS threshold. For the majority of proteins, family-specific ECIF::RF models generally exceeded the Generic AEV-PLIG::GATv2 baseline, supporting the hypothesis that model complexity should be matched to available data.

The results suggest an intermediate balance between retaining enough data to avoid overfitting and filtering enough data to preserve target-relevant information. This trend is most clearly observed for SYK, PFKFB3, HIF2α, and MAPK14, where Kτ peaked at a structural similarity threshold of approximately 0.3 to 0.5 before declining at higher thresholds as training sets became too small to support reliable models. At lower thresholds, substantially more training data were retained but performance often decreased relative to the intermediate thresholds, suggesting that structurally dissimilar data may dilute the target-relevant signal. Notably, the generic ECIF::RF model (threshold = 0.0) was not the highest-performing configuration for any of the five proteins.

Family-specific performance relative to the generic ECIF::RF baseline varied substantially across proteins. MCL1 showed a monotonically increasing trend with stricter filtering, with Kτ rising from 0.387 (generic ECIF::RF) to 0.527 at the target-specific level, a 36% improvement using only 0.27% of the full training database. This likely reflects its well-defined binding site (druggability score 0.91, cliff fraction 0.00; Figure 2C) and strong representation in the training data. PFKFB3 showed the most dramatic relative improvement, with Kendall's increasing from 0.141 (generic) to 0.392 at a threshold of 0.3, a 178% improvement using 3.2% of the full training database, providing the clearest evidence that data relevance can substantially outweigh data quantity. SYK showed a peak at threshold 0.3 (Kendall's = 0.428) before declining, consistent with the intermediate optimum pattern, though performance did not consistently exceed the Generic AEV-PLIG::GATv2 baseline across all thresholds. HIF2α showed less consistent behavior, likely reflecting the limited availability of structurally relevant training data at higher thresholds. MAPK14 showed a modest peak at threshold 0.4 (Kendall's = 0.537), with the relatively small magnitude of improvement potentially reflecting the high structural conservation of kinase binding sites, which limits the discriminative value of broad family-level filtering.

Figure 5 shows predicted versus experimental ΔG scatterplots for Generic AEV-PLIG::GATv2, Target-specific ECIF::RF, and the best family-specific ECIF::RF model across all five proteins. These plots highlight an important distinction between absolute prediction accuracy and ranking performance. Generic AEV-PLIG::GATv2 generally follows the overall experimental affinity trend more closely, while ECIF::RF models consistently show a compressed predicted affinity range. Despite this, ECIF::RF achieves stronger Kτ and Overlap@5 for four of five proteins, indicating better preservation of relative compound ordering even when absolute prediction accuracy is lower.

PFKFB3 provides the clearest illustration of this distinction (panels c, h, m). Generic AEV-PLIG::GATv2 achieves the lowest RMSE of 1.135 kcal/mol but recovers none of the five strongest binders (Overlap@5 = 0.00), whereas the best family-specific ECIF::RF model (FS 0.3) has a substantially higher RMSE of 1.760 kcal/mol yet correctly identifies two of the five strongest binders (Overlap@5 = 0.40) despite a visibly flattened prediction range. This illustrates that for lead optimization, where identifying the strongest binders for experimental follow-up is the primary goal, RMSE alone is a misleading measure of model utility. The exception is SYK (panels b, g, l), where Target-specific ECIF::RF shows degraded Overlap@5 relative to Generic AEV-PLIG::GATv2, reflecting the difficulty of this target when training data are severely limited. Performance partially recovers at the family-specific level, consistent with the intermediate optimum observed in Figure 4.

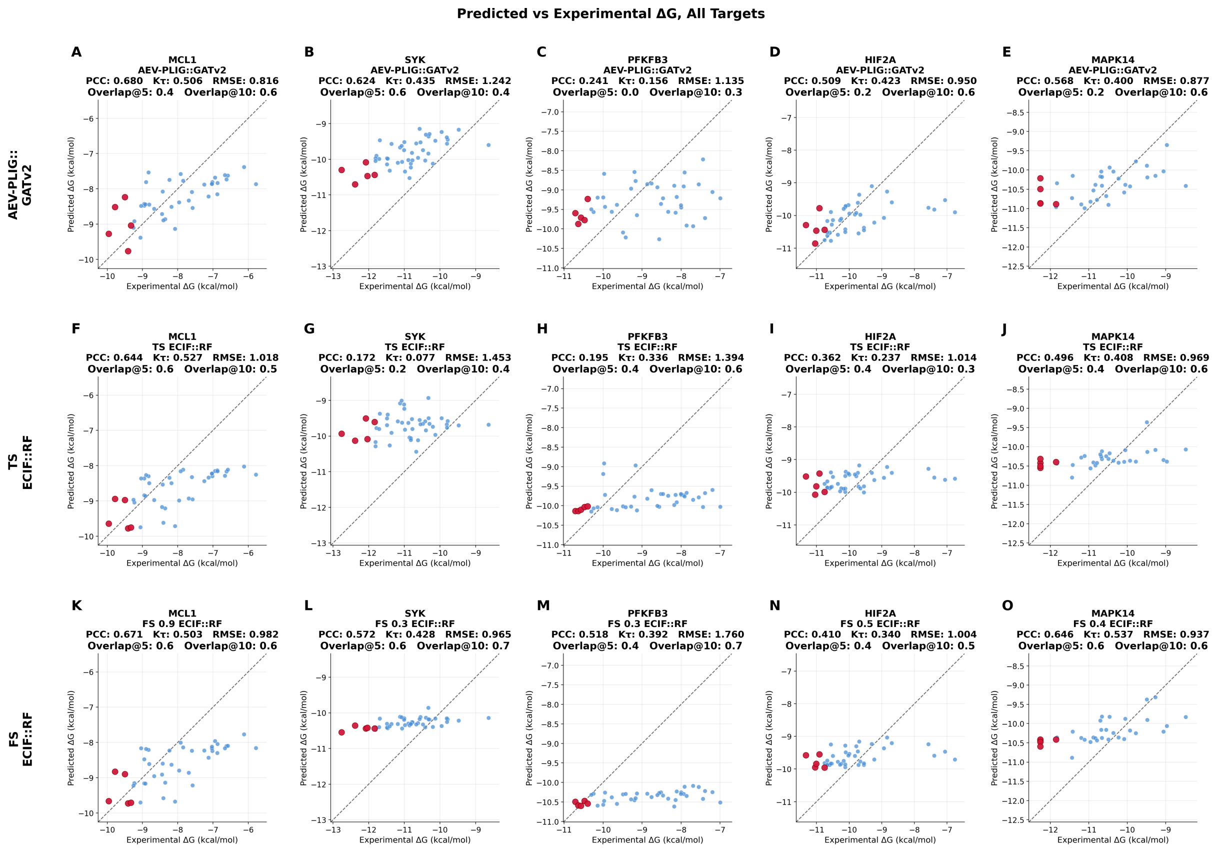


Figure 5: Predicted vs. Experimental ΔG for All Targets. (a–e) Generic AEV-PLIG::GATv2, (f–j) Target-specific ECIF::RF, and (k–o) best family-specific ECIF::RF models, for MCL1, SYK, PFKFB3, HIF2α, and MAPK14 respectively. Red points indicate the top-5 experimentally strongest binders; Overlap@5 is annotated in each panel. The dashed diagonal represents perfect prediction. Reading down each column, Generic AEV-PLIG::GATv2 tends to follow the overall affinity trend but frequently misses the strongest binders, while ECIF::RF models show a compressed predicted affinity range but improved Overlap@5 for four of five proteins. The exception is SYK (g), where Target-specific ECIF::RF shows degraded Overlap@5 relative to the generic baseline, reflecting the difficulty of learning from a severely limited target-specific training set; performance partially recovers at the family-specific level (l). The best family-specific threshold was selected by Kendall's on the test set and varies by protein: FS 0.9 for MCL1, FS 0.3 for SYK and PFKFB3, FS 0.5 for HIF2α, and FS 0.4 for MAPK14. This selection is optimistic and intended for illustrative comparison.

Figure 6 provides an overall summary comparison of all models at their best-performing threshold alongside FEP+ as an upper performance reference. The best family-specific threshold was selected by Kendall's on the test set for each model independently, making these point estimates optimistic; they are intended for illustrative comparison rather than definitive ranking.

Across proteins, the best family-specific ECIF models were competitive with or exceeded Generic AEV-PLIG::GATv2 in Kendall's for four of five proteins (MCL1, SYK, PFKFB3, MAPK14), with HIF2α as the exception where Generic AEV-PLIG::GATv2 retained a higher point estimate. All models remained substantially below FEP+ performance across all proteins, consistent with the known accuracy advantage of physics-based alchemical methods. The gap between ML scoring functions and FEP+ was largest for SYK and smallest for MAPK14, potentially reflecting the difficulty of SYK's ambiguous binding site and the abundance of kinase-family training data available for MAPK14.

Target-specific AEV-PLIG::GATv2 consistently underperformed its generic counterpart across all proteins, confirming that high-capacity GNN architectures are poorly suited to the small training sets available in the target-specific regime. In contrast, Target-specific ECIF::RF remained competitive with the generic AEV-PLIG::GATv2 baseline for three of five proteins (MCL1, PFKFB3, MAPK14), further supporting the suitability of lower-complexity models when training data are limited.

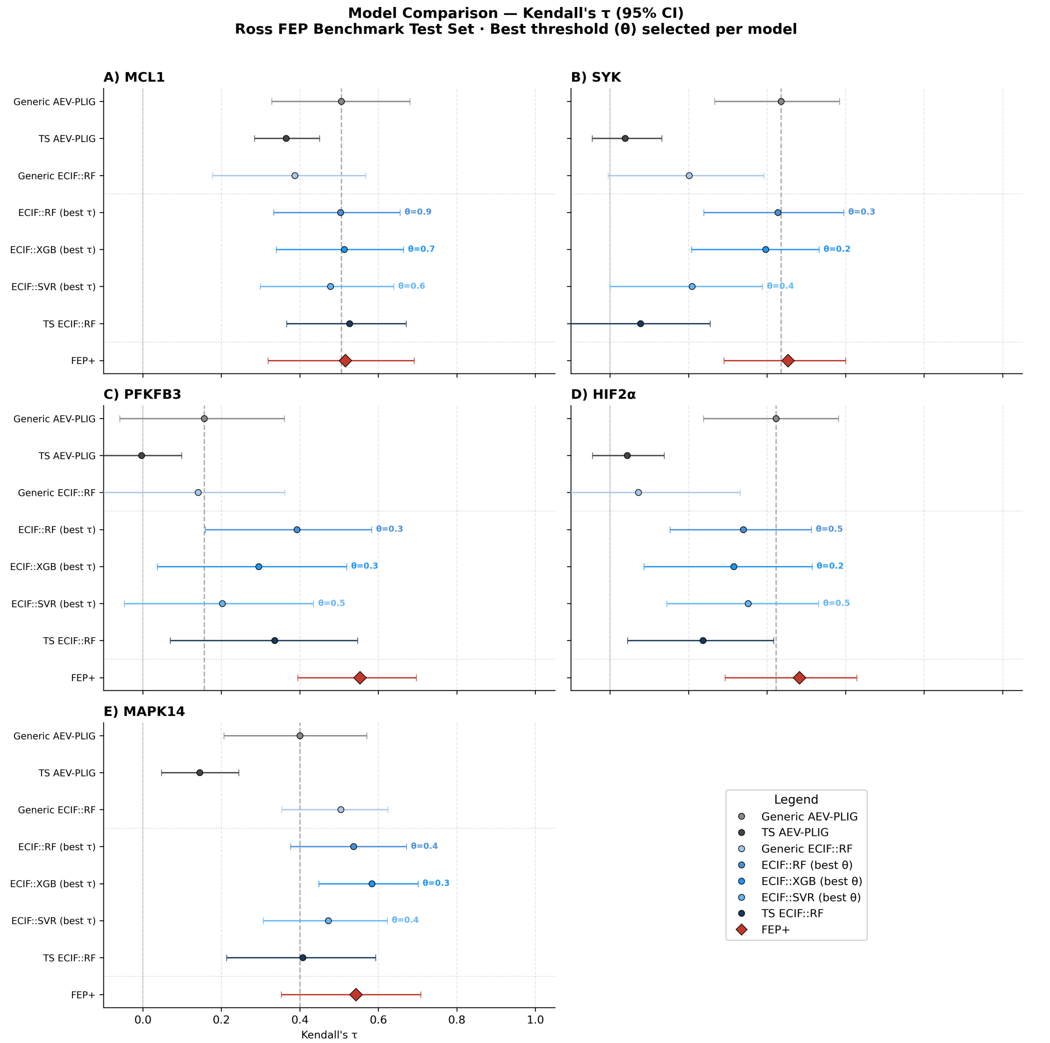


Figure 6: Model Comparison Summary. Forest plot showing Kendall's τ for all models evaluated on the Ross FEP Benchmark test set for each protein: (A) MCL1, (B) SYK, (C) PFKFB3, (D) HIF2α, and (E) MAPK14. Each panel shows Generic and Target-specific AEV-PLIG::GATv2, Generic and Target-specific ECIF::RF, and the best family-specific threshold for ECIF::RF, ECIF::XGB, and ECIF::SVR. FEP+ is shown as an upper performance reference representing a widely adopted physics-based alchemical relative binding free energy method. The highest performing (on Kτ) family-specific threshold per model is shown. Error bars represent 95% bootstrap confidence intervals. Wide confidence intervals reflect the small congeneric series sizes (n=34-46 compounds per protein) and confirm that all comparisons should be treated as exploratory.

Discussion

Binding affinity optimization is a complex, multi-parameter problem and an important application of CADD, particularly during lead optimization where the protein target and a promising scaffold have already been identified. At this stage, a fast model optimized for a specific target could be particularly useful. However, while training-set quantity is commonly emphasized in BAP, the relevance of that data to the target has received comparatively less attention. In this study, I investigated the tradeoff between data relevance and quantity across five protein targets, three featurization schemes, and four learning algorithms. Overall, the results suggest that more training data are not necessarily better when additional data are not sufficiently relevant to the target. Instead, there appears to be a balance between retaining enough structural diversity to avoid overfitting and filtering enough data to preserve target-relevant information.

Strong binders are often prioritized for experimental follow-up [6], meaning that correctly identifying the highest-affinity compounds can be more practically useful than minimizing global prediction error. This distinction is particularly important for lead optimization, where the practical goal is often to identify a small number of promising compounds rather than perfectly reproduce every experimental affinity. A model that compresses the predicted affinity range but correctly ranks the strongest compounds may therefore be more useful than a model with better absolute accuracy but poor ordering. Consequently, BAP models should not necessarily be evaluated solely through RMSE or PCC when the intended application is lead optimization. In this case, ranking and top-hit retrieval provide complementary measures of practical utility. However, the wide bootstrap confidence intervals, particularly for Overlap@5, mean that these differences should be considered exploratory rather than statistically definitive. The main finding is therefore not that a particular similarity threshold is universally optimal, but that data relevance can meaningfully influence BAP performance and should be considered alongside dataset size when constructing training sets.

ECIF::RF outperforms Generic AEV-PLIG::GATv2 in Overlap@5 and ranking performance, and consistently outperforms Target-specific and Family-specific AEV-PLIG::GATv2 across all data regimes. In low-data contexts, this is expected. Random Forest models are inherently regularized through bagging and feature subsampling, making them resistant to overfitting even with small training sets. GNNs, on the other hand, have far more capacity than can be supported by a small training set and are prone to overfitting. This can be interpreted as a complex manifestation of the bias-variance tradeoff, where model complexity must be balanced against the available data. Notably, hyperparameter ablation for AEV-PLIG::GATv2 resulted in an optimal configuration of 23M parameters compared to 8M in the original architecture, a substantially larger model that may have further exacerbated overfitting in the target-specific regime. This is consistent with observations in related virtual screening tasks, where simpler models were competitive with or outperformed deep learning approaches in top-hit retrieval when training data were limited [45]. Although the problem differs from BAP, both studies suggest that greater model complexity is not automatically beneficial when training data are limited. The results here extend this idea by showing that the relevance of those data may also determine which model class is most appropriate.

The majority of ECIF models outperform their PLEC counterparts, which likely reflects fundamental differences in the two featurization schemes. ECIF encodes protein-ligand interactions as a count vector of atom-pair types within a 6.0 Å cutoff, capturing a broad chemical environment around the binding site. PLEC uses circular topological hashing within a 4.5 Å cutoff, producing a higher-dimensional but more contact-focused representation. The broader interaction radius of ECIF may be more informative for the patterns necessary to predict binding affinity within the congeneric series of the Ross FEP test set. Within ECIF, Random Forest was the strongest learning algorithm, consistent with findings in the original ECIF paper [37].

An intermediate structural similarity threshold of approximately 0.3 to 0.5 was optimal for four of five proteins, suggesting a balance between training set diversity and target specificity. At low thresholds, the training distribution is dominated by structurally unrelated proteins whose SAR patterns are irrelevant or misleading for the target. At high thresholds, the training set becomes too small to reliably estimate the feature-affinity relationship. The optimum represents the point where the signal-to-noise ratio in the training data is maximized. This is related to the concept of co-variate shift, where a mismatch between the input distributions of training and test data degrades model performance [46]. At low thresholds, the training set contains proteins whose structural and chemical contexts differ substantially from the target, introducing a distributional mismatch that the model cannot overcome regardless of dataset size.

The results varied substantially between proteins, demonstrating that the optimal training strategy is target-dependent. MCL1 shows strong and consistent performance, potentially reflecting its relatively strong representation in the available training data and well-defined binding site. PFKFB3 and HIF2α show the greatest relative benefit from target- and family-specific training, making it the clearest example of the value of relevant data. PFKFB3 and HIF2α show the greatest relative benefit from relevance filtering within ECIF::RF, both showing substantial improvement from generic to best family-specific threshold. While PFKFB3's family-specific models also exceed Generic AEV-PLIG::GATv2, HIF2α does not, likely reflecting PFKFB3's more structurally distinct fold for which a generic model learns a particularly poor prior. This is particularly notable given that PFKFB3 currently lacks approved inhibitors, with candidates remaining at preclinical stages [47]. The absence of established chemical matter and a restricted training set size suggests that for underrepresented targets, data relevance is even more critical than quantity. Furthermore, the majority of structurally dissimilar sequences may encode SAR patterns that do not generalize to this non-conserved protein fold, such that the generic model learns a misleading prior. This is reminiscent of few-shot learning, where models trained on very small but highly relevant datasets can outperform those trained on larger but less specific source [26].

SYK shows a trend toward improved performance around a similarity threshold of 0.3, although performance did not consistently exceed the Generic AEV-PLIG::GATv2 baseline. For kinases such as SYK and MAPK14, binding sites are highly conserved across over 500 human kinases, selectivity may matter as much as affinity [48]. A model optimized purely for binding strength risks prioritizing compounds that also bind off-targets within the kinase AVOIDOME, a panel of proteins associated with toxicity and adverse effects [6]. MAPK14 provides a strong example of this challenge: despite decades of drug discovery efforts, no inhibitors have successfully completed clinical trials, with failures attributed primarily to toxicity and selectivity issues arising from the high conservation of kinase binding sites [49**?**]. This suggests that for kinases, optimizing for binding affinity alone is insufficient. Future work could extend the target-specific framework developed here to explicitly model selectivity alongside affinity, training not just on the target of interest but on related anti-targets, enabling the identification of compounds that are both potent and selective.

This work is not without its limitations and possible improvements. Here, I describe three main limitations, which, in turn, inspire future work that may further the findings of this paper. First, this study would benefit from a more comprehensive test set. The benchmark used is a rigorous congeneric series evaluated with FEP+, a valuable and challenging benchmark, but the computational requirements of FEP+ mean that only small congeneric series are available. Alternative benchmarks such as CASF-2016 [50] or the out-of-distribution (OOD) test set [20] could be explored, though isolating specific proteins from these sources presents its own difficulties. The small size of the current congeneric series limits the ability to draw statistically significant conclusions, and the narrow range of protein characteristics across the five targets means that findings are currently quite protein-dependent. Expanding to more proteins would result in more robust and transferable conclusions. Second, this study would benefit from a broader range of model types. AEV-PLIG::GATv2 is not representative of graph-based featurizations as a whole, and other GNN architectures should be explored before drawing general conclusions about this class of models. Additionally, further featurization specifications could be investigated for promising fingerprint representations, such as multi-shelled ECIF. Third, while this study already includes several rigorous methodological choices, such as split strategy evaluation, hyperparameter tuning, and Tanimoto similarity filtering, many recent machine learning papers call for stricter methodology to prevent models from memorizing training data. In this context, a stricter Tanimoto similarity cutoff between training and test sets could be applied. Optuna is used for hyperparameter tuning in the fingerprint-based models but not in AEV-PLIG::GATv2, which could be streamlined in future work. More rigorous analysis of model overfitting and learning curve analysis would also be beneficial. Beyond these limitations, the results open the door to several directions for future work, including transfer learning, fine-tuning, isolation of the training set from the augmented database, and additional docking steps. These approaches could help define an optimal method for combining and balancing generic models with target-specific models, and outline a more robust and generalizable pipeline for training target-optimized models for any protein of interest.

This study demonstrates that data relevance is an important and relatively underexplored axis in binding affinity prediction for target-specific lead optimization. The results suggest that increasing training-set size does not necessarily improve performance when additional data are not sufficiently relevant to the target. Instead, there appears to be a tradeoff between data quantity, relevance, and model complexity. The substantial target-to-target differences and wide confidence intervals mean that these findings should be treated as exploratory. Rather than demonstrating that one model or similarity threshold is universally optimal, this work motivates a broader consideration of training-data selection in BAP. In essence, we are not presenting a model that is simply “better”, but an alternative route that could benefit the way BAP is approached in the long run. A faster and more intentional target-specific model will never have the breadth of a computationally expensive GNN or an alchemical physics-based method, but it may provide the best route for a model suited to its specific purpose. ECIF::RF is substantially faster than AEV-PLIG::GATv2 and FEP+ [20], providing a practical advantage for iterative lead optimization where models may need to be repeatedly retrained as new experimental data become available.

References

1. O. J. Wouters, M. McKee, J. Luyten, Estimated Research and Development Investment Needed to Bring a New Medicine to Market, 2009-2018. *JAMA* **323**, 844 (2020).
2. I. Kapetanovic, COMPUTER-AIDED DRUG DISCOVERY AND DEVELOPMENT (CADDD): in silico-chemico-biological approach. *Chemico-biological interactions* **171**, 165 (2008).
3. G. Sliwoski, S. Kothiwale, J. Meiler, E. W. Lowe, Computational Methods in Drug Discovery. *Pharmacological Reviews* **66**, 334 (2014).
4. L. G. Ferreira, R. N. Dos Santos, G. Oliva, A. D. Andricopulo, Molecular Docking and Structure-Based Drug Design Strategies. *Molecules* **20**, 13384 (2015).
5. M. Aldeghi, V. Gapsys, B. L. de Groot, Accurate Estimation of Ligand Binding Affinity Changes upon Protein Mutation. *ACS Central Science* **4**, 1708 (2018).
6. M. A. Murcko, The Affinity Advantage. *Journal of Medicinal Chemistry* **69**, 1963 (2026).
7. Y. Momin, V. Beloshe, Pharmacophore modeling in drug design. *Advances in Pharmacology* **103**, 313

(2025).

8. L. Wang, *et al.*, Accurate and Reliable Prediction of Relative Ligand Binding Potency in Prospective Drug Discovery by Way of a Modern Free-Energy Calculation Protocol and Force Field. *Journal of the American Chemical Society* **137**, 2695 (2015).
9. R. Abel, L. Wang, E. D. Harder, B. J. Berne, R. A. Friesner, Advancing Drug Discovery through Enhanced Free Energy Calculations. *Accounts of Chemical Research* **50**, 1625 (2017).
10. H. Green, D. R. Koes, J. D. Durrant, DeepFrag: a deep convolutional neural network for fragment-based lead optimization. *Chemical Science* **12**, 8036 (2021).
11. Deep lead optimization enveloped in protein pocket and its application in designing potent and selective ligands targeting LTK protein | Nature Machine Intelligence.
12. P. J. Ballester, A. Schreyer, T. L. Blundell, Does a More Precise Chemical Description of Protein–Ligand Complexes Lead to More Accurate Prediction of Binding Affinity? *Journal of Chemical Information and Modeling* **54**, 944 (2014).
13. J. Iwakiri, M. Hamada, K. Asai, T. Kameda, Improved Accuracy in RNA–Protein Rigid Body Docking by Incorporating Force Field for Molecular Dynamics Simulation into the Scoring Function. *Journal of Chemical Theory and Computation* **12**, 4688 (2016).
14. I. A. Guedes, F. S. S. Pereira, L. E. Dardenne, Empirical Scoring Functions for Structure-Based Virtual Screening: Applications, Critical Aspects, and Challenges. *Frontiers in Pharmacology* **9**, 1089 (2018).
15. J. H. Lam, V. Katritch, Navigating structure-based drug discovery with emerging innovations in physics- and knowledge-based approaches. *npj Drug Discovery* **2**, 29 (2025).
16. S.-Y. Huang, S. Z. Grinter, X. Zou, Scoring functions and their evaluation methods for protein-ligand docking: recent advances and future directions. *Physical chemistry chemical physics: PCCP* **12**, 12899 (2010).
17. H. Li, K.-H. Sze, G. Lu, P. J. Ballester, Machine-learning scoring functions for structure-based drug lead optimization. *WIREs Computational Molecular Science* **10**, e1465 (2020). _eprint: https://wires.onlinelibrary.wiley.com/doi/pdf/10.1002/wcms.1465.
18. G. A. Ross, *et al.*, The maximal and current accuracy of rigorous protein-ligand binding free energy calculations. *Communications Chemistry* **6**, 222 (2023).
19. I. Muegge, Y. Hu, Recent Advances in Alchemical Binding Free Energy Calculations for Drug Discovery. *ACS Medicinal Chemistry Letters* **14**, 244 (2023).
20. □. Valsson, *et al.*, Narrowing the gap between machine learning scoring functions and free energy perturbation using augmented data. *Communications Chemistry* **8**, 41 (2025).
21. R. Meli, G. M. Morris, P. C. Biggin, Scoring Functions for Protein-Ligand Binding Affinity Prediction Using Structure-based Deep Learning: A Review. *Frontiers in Bioinformatics* **2** (2022).
22. B. Ramsundar, *et al.*, Massively Multitask Networks for Drug Discovery. (2015). ArXiv:1502.02072 [stat.ML].
23. D. L. Mobley, M. K. Gilson, Predicting binding free energies: Frontiers and benchmarks. *Annual review of biophysics* **46**, 531 (2017).
24. D. Stumpfe, J. Bajorath, Exploring Activity Cliffs in Medicinal Chemistry. *Journal of Medicinal Chemistry* **55**, 2932 (2012).
25. D. van Tilborg, A. Alenicheva, F. Grisoni, Exposing the Limitations of Molecular Machine Learning with Activity Cliffs. *Journal of Chemical Information and Modeling* **62**, 5938 (2022).
26. Low Data Drug Discovery with One-Shot Learning | ACS Central Science.
27. Protein–Ligand Scoring with Convolutional Neural Networks | Journal of Chemical Information and Modeling.
28. M. A. Lones, Avoiding common machine learning pitfalls. *Patterns* **5** (2024).
29. A Comparative Assessment of Predictive Accuracies of Conventional and Machine Learning Scoring Functions for Protein-Ligand Binding Affinity Prediction.
30. Z. Liu, *et al.*, Forging the Basis for Developing Protein–Ligand Interaction Scoring Functions. *Accounts of Chemical Research* **50**, 302 (2017).
31. X. Li, *et al.*, A High-Quality Data Set of Protein–Ligand Binding Interactions Via Comparative Complex Structure Modeling. *Journal of Chemical Information and Modeling* **64**, 2454 (2024).
32. M. K. Gilson, *et al.*, BindingDB in 2015: A public database for medicinal chemistry, computational chemistry and systems pharmacology. *Nucleic Acids Research* **44**, D1045 (2016).
33. D. Rogers, M. Hahn, Extended-Connectivity Fingerprints. *Journal of Chemical Information and Modeling* **50**, 742 (2010).
34. D. Butina, Unsupervised Data Base Clustering Based on Daylight's Fingerprint and Tanimoto Similarity: A Fast and Automated Way To Cluster Small and Large Data Sets. *Journal of Chemical Information and Computer Sciences* **39**, 747 (1999).
35. G. W. Bemis, M. A. Murcko, The Properties of Known Drugs. 1. Molecular Frameworks. *Journal of Medicinal Chemistry* **39**, 2887 (1996).
36. Y. Zhang, J. Skolnick, TM-align: a protein structure alignment algorithm based on the TM-score. *Nucleic Acids Research* **33**, 2302 (2005).
37. N. Sánchez-Cruz, J. L. Medina-Franco, J. Mestres, X. Barril, Extended connectivity interaction features: improving binding affinity prediction

through chemical description. *Bioinformatics* **37**, 1376 (2021).
38. M. Wójcikowski, M. Kukiełka, M. M. Stepniewska-Dziubinska, P. Siedlecki, Development of a protein–ligand extended connectivity (PLEC) fingerprint and its application for binding affinity predictions. *Bioinformatics* **35**, 1334 (2019).
39. S. Brody, U. Alon, E. Yahav (2021).
40. L. Breiman, Random Forests. *Machine Learning* **45**, 5 (2001).
41. T. Chen, C. Guestrin, *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, KDD '16 (Association for Computing Machinery, New York, NY, USA, 2016), pp. 785–794.
42. H. Drucker, C. J. C. Burges, L. Kaufman, A. Smola, V. Vapnik, *Advances in Neural Information Processing Systems* (MIT Press, 1996), vol. 9.
43. [1907.10902] Optuna: A Next-generation Hyperparameter Optimization Framework.
44. V. Le Guilloux, P. Schmidtke, P. Tuffery, Fpocket: An open source platform for ligand pocket detection. *BMC Bioinformatics* **10**, 168 (2009).
45. C.-M. Truong, P. J. Ballester, O. Taboureau, V.-K. Tran-Nguyen, NextTopDocker: the largest-to-date docking power benchmark reveals that deep learning performs generally much worse than logistic regression models. (2026).
46. M. Sugiyama, M. Kawanabe, *Machine Learning in Non-Stationary Environments: Introduction to Covariate Shift Adaptation* (The MIT Press, 2012).
47. R. Bartrons, *et al.*, The potential utility of PFKFB3 as a therapeutic target. *Expert Opinion on Therapeutic Targets* **22**, 659 (2018).
48. M. W. Karaman, *et al.*, A quantitative analysis of kinase inhibitor selectivity. *Nature Biotechnology* **26**, 127 (2008).
49. M. M. Madkour, H. S. Anbar, M. I. El-Gamal, Current status and future prospects of p38α/MAPK14 kinase and its inhibitors. *European Journal of Medicinal Chemistry* **213**, 113216 (2021).
50. M. Su, *et al.*, Comparative Assessment of Scoring Functions: The CASF-2016 Update. *Journal of Chemical Information and Modeling* **59**, 895 (2018).
51. A. N. Jain, Surflex: Fully Automatic Flexible Molecular Docking Using a Molecular Similarity-Based Search Engine. *Journal of Medicinal Chemistry* **46**, 499 (2003).
52. P. Veličković, *et al.*, Graph Attention Networks. (2018). ArXiv:1710.10903 [stat.ML].

Abbreviations

ADME, Absorption Distribution Metabolism Excretion;
AEV, Atomic Environment Vector;
BAP, Binding Affinity Prediction;
BFE, Binding Free Energy;
BindingDB-DCS, Binding database docked congeneric series;
BLAST, Basic Local Alignment Search Tool;
CADD, Computer-Aided Drug Design;
ECFP6, Extended Connectivity Fingerprint 6;
ECIF, Extended Connectivity Interaction Fingerprints;
FEP, Free Energy Perturbation;
FS, Family-specific;
GNN, Graph Neural Network;
GT, Ground Truth;
HIF2α, Hypoxia-Inducible Factor 2α;
HTS, High-Throughput Screening;
Kτ, Kendall's tau;
LBVS, Ligand-based Virtual Screening;
MAPK14, p38 MAP Kinase;
MCL1, Myeloid Cell Leukemia 1;
MMseqs2, Many-against-Many sequence searching;
OOD, out-of-distribution;
PCC, Pearson correlation coefficient;
PDB, Protein Data Bank;
PFKFB3, Phosphofructokinase Fructose Bisphosphatase 3;
PLEC, Protein Ligand Extended Connectivity;
PLIG, Protein Ligand Interaction Graph;
QSAR, Quantitative Structure-Activity Relationship;
RF, Random Forest;
RMSE, root mean square error;
SAR, Structure activity relationship;
SBVS, Structure-based Virtual Screening;
SVR, Support Vector Regression;
SYK, Spleen Tyrosine Kinase;
TM-align, Template Modeling alignment;
TPE, Tree-structured Parzen Estimator;
TS, Target-specific;
VS, Virtual Screening;
XGB, Extreme Gradient Boosting

Supplementary Methods

Training Databases

PDBbind v2020 is partitioned into a refined set (5,316 complexes) and a general set (14,127 complexes). The refined subset satisfies more stringent criteria, including a minimum crystal structure resolution of 2.5 Å and only K_i or K_d affinity measurements. Preprocessing with RDKit v2023 removed 135 complexes containing ligands with uncommon elements (i.e. not H, B, C, N, O, F, P, S, Cl, Br, or I) or unknown bond types, yielding 19,308 complexes for training. BindingNet [31] was constructed by identifying candidate ligands from ChEMBL with binding activity against PDBbind v2019 targets. For each protein–ligand pair, 3D poses were generated via template-based alignment using the maximum common substructure (MCS) between candidate and reference crystal ligands. RDKit preprocessing removed 89 ligands containing unsupported elements, yielding 69,727 complexes spanning 802 unique targets.BindingDB-DCS [32] consists of congeneric ligand series with experimentally measured binding affinities and docking-derived poses generated using Surflex-Dock [51]. Congeneric series were identified via Tanimoto similarity from ChemAxon fingerprints. Entries with censored affinity values (e.g., ">" or "<" annotations) and ligands unparseable by RDKit were removed. Where multiple affinity types were available, K_d was prioritized over K_i and K_i over IC$_{50}$, yielding 8,822 complexes spanning 1,311 unique protein structures.

AEV-PLIG

AEV-PLIG featurization captures local atomic chemical environments using radial symmetry functions; in this implementation, only radial features are used. For ligand atom i, the scalar entry $g_i^{t,m}$ of the radial AEV vector G_i^{Rad} is indexed by protein atom type t and parameter pair $m = (\eta, R_s)$, and computed as a sum over all protein atoms j of atom type t:

$$g_i^{t,m} = \sum_{j \in \mathcal{I}_{\text{protein},\,t}} e^{-\eta(R_{ij}-R_s)^2} f_c(R_{ij}) \tag{1}$$

where f_c is a smooth cutoff function that enforces a finite interaction radius R_c:

$$f_c(R_{ij}) = \begin{cases} \frac{1}{2}\left[\cos\left(\frac{\pi R_{ij}}{R_c}\right) + 1\right], & R_{ij} \le R_c, \\ 0, & R_{ij} > R_c. \end{cases} \tag{2}$$

Entries are computed for all combinations of $t \in \mathcal{T}$ and $m = (\eta, R_s)$, yielding a descriptor vector of length $N \times L \times K = 22 \times 16 \times 1 = 352$ per ligand atom. Default parameters are adopted from Valsson et al. [20]: $\eta = 19.7$ and $R_c = 5.1$ Å. AEV descriptors are computed for each atom in the protein–ligand complex and used as node features in the GATv2 graph neural network. Edge features encode the interatomic distance R_{ij} between connected atoms, represented as a scalar value within the cutoff radius R_c.

GATv2

GATv2 [39] is an attention graph neural network and a modified version of GATs [52]. The updated feature vector for node u is defined in Equation (3), computed as a weighted average of the linearly transformed features of neighboring nodes after the application of a nonlinearity σ.

$$h_u = \sigma\left(\sum_{v \in \mathcal{N}_u \cup \{u\}} a_{uv} \mathbf{W}_t x_v\right) \tag{3}$$

The attention coefficient a_{uv} between nodes u and v is defined in Equation 4: where $\mathbf{W}_s$, $\mathbf{W}_t$, and $\mathbf{W}_e$ are learnable weight matrices for the source node, target node, and edge features respectively, **b** is a learnable attention vector, e_{uv} denotes the edge feature between nodes u and v, and $\mathcal{N}_u \cup \{u\}$ denotes the neighborhood of node u including itself.

$$a_{uv} = \frac{\exp\left(\mathbf{b}^\top \text{LeakyReLU}\left(\mathbf{W}_s x_u + \mathbf{W}_t x_v + \mathbf{W}_e e_{uv}\right)\right)}{\sum_{k \in \mathcal{N}_u \cup \{u\}} \exp\left(\mathbf{b}^\top \text{LeakyReLU}\left(\mathbf{W}_s x_u + \mathbf{W}_t x_k + \mathbf{W}_e e_{u,k}\right)\right)} \tag{4}$$

GATv2 hyperparameters are reported in Supplementary Figure 4 and available at `https://github.com/chyiricketts/Target-Specific_BAP_Prediction`.

$$\hat{y}_i = \phi(\mathbf{x}_i) = \sum_{k=1}^{K} f_k(\mathbf{x}_i), \quad f_k \in \mathcal{F} \tag{5}$$

where $\mathcal{F}$ is the space of regression trees and f_k is the k-th tree function.

Performance Metrics

Model performance is evaluated using four complementary metrics. Pearson Correlation Coefficient (PCC) measures the linear correlation between experimental and predicted binding free energies ΔG:

$$\text{PCC} = \frac{\sum_{i=1}^{n}(\hat{y}_i - \bar{\hat{y}})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(\hat{y}_i - \bar{\hat{y}})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{6}$$

where y_i are experimental ΔG values, $\hat{y}_i$ are predicted values, and $\bar{y}$, $\bar{\hat{y}}$ are their respective means. PCC ranges from -1 to 1, with higher values indicating stronger linear agreement.

Kendall's τ ($K\tau$) measures rank concordance between predicted and experimental ΔG:

$$K\tau = \frac{n_c - n_d}{\frac{1}{2}n(n-1)} \tag{7}$$

where n_c and n_d are the numbers of concordant and discordant pairs respectively, and n is the total number of compounds. $K\tau$ is particularly relevant for lead optimization, where correct compound ranking is often more important than absolute affinity prediction.

Root Mean Square Error (RMSE) measures the average magnitude of prediction errors in kcal/mol:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2} \tag{8}$$

where lower values indicate more accurate absolute predictions.

Top-K overlap (Overlap@K) computes the percentage overlap between the top-K strongest binders identified by the model and the ground truth (GT) derived from the experimental values. Overlap@K is reported as a proportion in the range [0, 1], where 1 indicates perfect retrieval of all top-K compounds:

$$\text{Overlap@K} = \frac{|\text{Top-K}_{\text{model}} \cap \text{Top-K}_{\text{GT}}|}{k} \tag{9}$$

where strongest binders are considered matched if they are within the set of K. This metric provides a practically relevant measure of model utility in lead optimization, where identifying the highest-affinity compounds for experimental follow-up is the primary goal.

Statistical Testing

To assess whether predictions from method M_1 significantly outperform those of method M_2 with respect to a metric ρ (e.g. PCC or $K\tau$), a bootstrapping procedure was employed. For each of $B = 1{,}000$ bootstrap samples, compounds were resampled with replacement from the congeneric series and the difference in metric was computed:

$$\Delta\rho_b = \rho_b^{M_1} - \rho_b^{M_2} \tag{10}$$

The p-value is reported as the fraction of bootstrap samples for which $\Delta\rho_b < 0$:

$$p = \frac{1}{B}\sum_{b=1}^{B}\mathbf{1}[\Delta\rho_b < 0] \tag{11}$$

A p-value < 0.05 is considered to indicate statistical significance. Given the small size of the congeneric test series ($n = 34$–46 compounds), wide confidence intervals are expected and most comparisons are treated as exploratory.

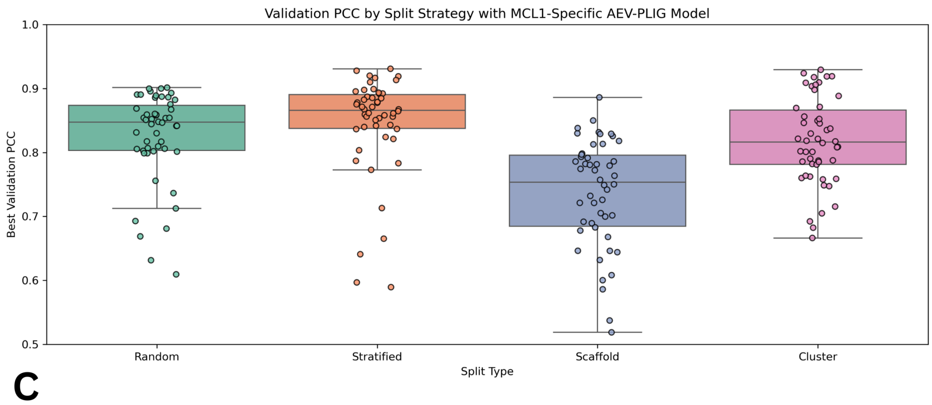
Additionally, pairwise bootstrap confidence intervals on metric differences were computed by resampling shared compound indices simultaneously for both models, with results reported in Supplementary Tables 6 and 7.

Supplementary Figures and Tables

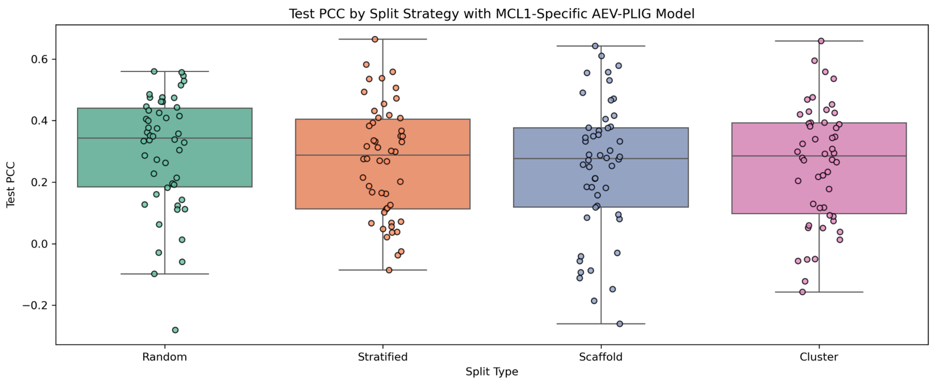
A

Split	Max Sim.	Mean > 0.8 Sim.	Mean > 0.9 Sim.
Random	1.00	21	3
Stratified	1.00	21	4
Scaffold	0.895	14	0
Cluster	0.884	2	0

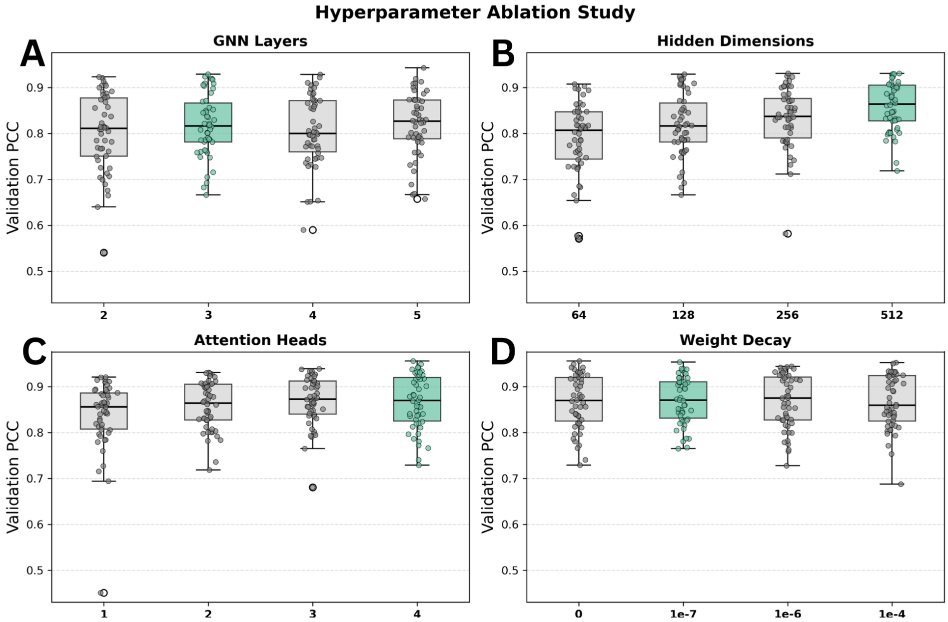
B



C



Supplementary Figure 1: Split Strategy Evaluation. (a) This table reports similarity between the training and validation sets as maximum similarity (Max Sim.) and mean similarity above 0.8 (Mean $\geq$ 0.8 Sim.) and 0.9 (Mean $\geq$ 0.9 Sim.) ECFP6 Tanimoto similarity. Random and stratified splits show near-identical compounds between train and validation (Max Sim. = 1.0) resulting in inflated Validation PCC. (b) MCL1-specific AEV-PLIG::GATv2 model Validation PCC by split strategy. Scaffold split over-separates, yielding unstable validation performance. Cluster split provides a principled balance between structural separation and distributional representativeness. *One outlier for scaffold split fold 5 is not shown at 0.24 validation PCC for visualization purposes. (c) MCL1-specific AEV-PLIG::GATv2 model Test PCC by split strategy. Test Set PCC is not used in split strategy selection, but instead shown for data transparency. Despite the use of a 5-fold cluster split, the gap between training and test performance (0.97–0.99 across configurations) suggests overfitting.



Supplementary Figure 2: Hyperparameter Ablation and Target-specific AEV-PLIG::GATV2 Test Set Performance. A hyperparameter ablation study was conducted to investigate optimal (a) GNN layers, (b) hidden dimensions, (c) attention heads, and (d) weight decay of target-specific AEV-PLIG::GATv2 models on cluster split, starting from a baseline of 3 GNN layers, 128 hidden dimensions, 2 attention heads, and no weight decay, with each subplot representing the Validation PCC of each model when modifying the specified variable. MCL1-specific training set was used at it represents the median target-specific training set size. The optimal setup was determined to be: 3 GNN layers, 512 hidden dimensions, 4 attention heads, and $1e^{-7}$ weight decay. (e) Target-specific AEV-PLIG::GATv2 Test Set performance using determined optimal hyperparameters (box) compared to generic AEV-PLIG::GATv2 model performance on each congeneric Test Set (black dashed line). Each box represents an ensemble 50 models (5 cluster split folds x 10 seeds). This optimal model contains 23,197,185 parameters, compared to 7,554,561 parameters (256 hidden dimensions, 5 hidden layers, 3 attention heads) in the original architecture, representing a substantial increase in capacity. All hyperparameter tuning and split strategy investigation is evaluated by validation set metrics as to reduce the risk of biased selection and overfitting [28].

Model	Hyperparameter	Type	Range/Options	Scale
XGB	n_estimators	Integer	300–3000	Linear
	max_depth	Integer	3–10	Linear
	learning_rate	Float	0.01–0.4	Log
	subsample	Float	0.5–1.0	Linear
	colsample_bytree	Float	0.5–1.0	Linear
	min_child_weight	Integer	1–10	Linear
	reg_alpha	Float	0.0–1.0	Linear
	reg_lambda	Float	0.5–5.0	Linear
	gamma	Float	0.0–5.0	Linear
RF	n_estimators	Integer	100–1500	Linear
	max_depth	Integer	5–50	Linear
	min_samples_split	Integer	2–20	Linear
	min_samples_leaf	Integer	1–10	Linear
	max_features	Categorical	sqrt, log2, None	–
	bootstrap	Fixed	True	–
SVR	C	Float	0.1–100	Log
	epsilon	Float	0.01–1.0	Linear
	gamma	Categorical	scale, auto	–
	kernel	Fixed	rbf	–

Supplementary Table 3: Hyperparameter search space for XGBoost, Random Forest, and SVR models used in Optuna optimization. Selected hyperparameters for all target-specific and family-specific models are available at: https://github.com/chyiricketts/Target-Specific_BAP_Prediction

Protein	Model	Training Size	PCC	KT	RMSE	O@5	O@10	KT p 1	KT p 2
MCL1	generic	97,692	0.5034	0.3872	1.1392	0.6	0.5	0.068	–
	0.1	94,478	0.5220	0.3988	1.1294	0.6	0.5	0.081	0.776
	0.2	82,352	0.5428	0.3942	1.0918	0.6	0.6	0.061	0.581
	0.3	58,701	0.4984	0.3593	1.1400	0.6	0.5	0.033	0.312
	0.4	4,167	0.5570	0.3965	1.1581	0.6	0.6	0.065	0.570
	0.5	681	0.6249	0.4849	1.1635	0.6	0.6	0.371	0.891
	0.6	462	0.6209	0.4709	1.1520	0.6	0.6	0.262	0.844
	0.7	249	0.6374	0.4988	1.0386	0.6	0.6	0.448	0.933
	0.8	122	0.6612	0.4849	0.9905	0.6	0.6	0.385	0.885
	0.9	119	0.6709	0.5035	0.9823	0.6	0.6	0.481	0.914
	target-specific	267	0.6438	0.5267	1.0180	0.6	0.5	0.637	0.954
SYK	generic	97,686	0.2697	0.2013	1.2279	0.4	0.4	0.005	–
	0.1	88,974	0.1828	0.1103	1.2186	0.2	0.4	0.001	0.017
	0.2	71,837	0.4446	0.3425	1.0816	0.4	0.4	0.184	0.977
	0.3	34,492	0.5725	0.4277	0.9647	0.6	0.7	0.489	1.000
	0.4	18,498	0.4718	0.3154	1.0732	0.6	0.5	0.148	0.992
	0.5	15,231	0.4992	0.3774	1.0625	0.8	0.6	0.263	0.992
	0.6	11,432	0.4786	0.4045	1.0558	0.4	0.6	0.354	0.999
	0.7	10,978	0.5506	0.3948	1.1175	0.6	0.6	0.326	0.998
	0.8	7,788	0.5098	0.3193	1.0307	0.4	0.5	0.093	0.940
	0.9	754	0.3453	0.2458	0.8414	0.0	0.4	0.024	0.690
	target-specific	559	0.1720	0.0774	1.4526	0.2	0.4	0.000	0.140
pfkfb3	generic	97,726	0.1394	0.1410	1.2687	0.4	0.4	0.406	–
	0.1	80,921	0.4575	0.3385	1.3420	0.4	0.6	0.939	0.996
	0.2	31,260	0.2682	0.2410	1.6271	0.2	0.5	0.771	0.782
	0.3	3,097	0.5182	0.3923	1.7600	0.4	0.7	0.919	0.963
	0.4	14	0.2902	0.2502	1.4227	0.0	0.4	0.670	0.739
	0.5	12	0.2760	0.2128	1.5618	0.0	0.3	0.621	0.659
	0.6	12	0.2894	0.2308	1.5495	0.0	0.4	0.660	0.695
	0.7	12	0.2820	0.2205	1.5350	0.0	0.3	0.638	0.666
	0.8	12	0.3771	0.2795	1.5067	0.2	0.5	0.735	0.794
	0.9	12	0.2906	0.2410	1.5496	0.0	0.4	0.691	0.727
	target-specific	65	0.1949	0.3359	1.3941	0.4	0.6	0.890	0.941
HIF2α	generic	97,716	0.0217	0.0723	1.1504	0.0	0.2	0.007	–
	0.1	95,840	0.0175	0.0944	1.1664	0.2	0.2	0.006	0.689
	0.2	86,606	-0.0402	0.0282	1.1508	0.0	0.1	0.004	0.174
	0.3	65,880	0.3005	0.3176	1.0249	0.0	0.5	0.175	0.996
	0.4	2,682	0.3986	0.2538	0.9914	0.2	0.3	0.029	0.875
	0.5	87	0.4100	0.3397	1.0037	0.4	0.5	0.093	0.981
	0.6	68	0.3661	0.3029	1.0212	0.4	0.4	0.037	0.952
	0.7	64	0.3615	0.2489	1.0027	0.4	0.2	0.016	0.877
	0.8	51	0.2529	0.2072	1.0277	0.4	0.3	0.008	0.838
	target-specific	61	0.3616	0.2367	1.0138	0.4	0.3	0.008	0.869
MAPK14	generic	97,731	0.6298	0.5045	0.9085	0.6	0.6	0.857	–
	0.1	83,734	0.5895	0.4758	0.9463	0.6	0.5	0.761	0.221
	0.2	58,576	0.5803	0.4973	0.9242	0.6	0.6	0.796	0.435
	0.3	23,722	0.6335	0.5153	0.9756	0.2	0.5	0.871	0.604
	0.4	12,751	0.6460	0.5368	0.9371	0.6	0.6	0.908	0.748
	0.5	11,345	0.6005	0.4758	0.8901	0.4	0.5	0.769	0.327
	0.6	9,109	0.6329	0.4542	0.7909	0.4	0.5	0.692	0.191
	0.7	1,950	0.5960	0.4183	0.8365	0.6	0.5	0.595	0.070
	0.8	866	0.6132	0.3968	0.8692	0.6	0.5	0.509	0.065
	0.9	600	0.5688	0.4004	0.9102	0.6	0.5	0.505	0.139
	target-specific	1,713	0.4957	0.4076	0.9691	0.4	0.6	0.532	0.205

Supplementary Table 4: ECIF::RF results across proteins and training sets including: generic (0.0), target-specific, and family-specific thresholds. PCC, KT, RMSE, and Overlap@5 are taken from the selected-set metrics (no bootstrap resampling). Overlap@K is reported as Overlap@5. Both KT p-values are one-sided paired bootstrap comparisons (1000 resamples) on the same selected compounds: the first against the Generic AEV-PLIG reference model, the second against this same ECIF::RF family's own threshold-0.0 baseline which isolates target-specificity independent of architecture. $p < 0.05$ indicates the reference scored higher on Kτ in significantly more than half of paired resamples; $p > 0.95$ indicates this row's model scored significantly higher than the reference.

Protein	Model	Training Size	PCC	KT	RMSE	Overlap@5	KT p 1	KT p 2
mcl1	generic	97,692	0.5400	0.3616	1.0706	0.4000	0.033	–
	0.1	94,478	0.4773	0.2965	1.0764	0.6000	0.006	0.096
	0.2	82,352	0.5383	0.3477	1.0379	0.6000	0.019	0.402
	0.3	58,701	0.5271	0.3663	1.0272	0.6000	0.026	0.537
	0.4	4,167	0.5881	0.4012	0.9347	0.6000	0.062	0.702
	0.5	681	0.6040	0.4826	1.0130	0.6000	0.350	0.967
	0.6	462	0.6712	0.5012	0.9500	0.6000	0.476	0.979
	0.7	249	0.6583	0.5128	0.9086	0.6000	0.525	0.988
	0.8	122	0.6627	0.4965	0.8595	0.6000	0.422	0.968
	0.9	119	0.6831	0.5012	0.8031	0.6000	0.486	0.973
	target-specific	267	0.6151	0.4500	0.8708	0.4000	0.208	0.858
syk	generic	97,686	0.3456	0.2380	0.9313	0.2000	0.011	–
	0.1	88,974	0.4910	0.3445	0.9582	0.6000	0.117	0.946
	0.2	71,837	0.5703	0.3967	0.8163	0.4000	0.296	0.994
	0.3	34,492	0.4001	0.2767	0.8971	0.4000	0.052	0.681
	0.4	18,498	0.4394	0.3135	0.9077	0.2000	0.106	0.882
	0.5	15,231	0.2140	0.0987	0.9916	0.2000	0.003	0.071
	0.6	11,432	0.3776	0.2612	1.0300	0.2000	0.045	0.655
	0.7	10,978	0.3439	0.2380	1.0708	0.2000	0.020	0.516
	0.8	7,788	0.2815	0.2051	1.0509	0.2000	0.014	0.348
	0.9	754	0.3109	0.2109	0.7811	0.0000	0.014	0.367
	target-specific	559	0.3630	0.1626	1.2880	0.2000	0.004	0.236
pfkfb3	generic	97,726	0.1973	0.1410	1.2492	0.6000	0.436	–
	0.1	80,921	0.2813	0.2564	1.5391	0.4000	0.821	0.891
	0.2	31,260	0.2613	0.2026	1.8159	0.2000	0.672	0.719
	0.3	3,097	0.3525	0.2949	1.6624	0.4000	0.790	0.835
	0.4	14	0.2938	0.2563	1.4051	0.0000	0.721	0.740
	0.5	12	0.1834	0.2002	1.4674	0.0000	0.606	0.630
	0.6	12	0.2944	0.1917	1.4355	0.0000	0.565	0.588
	0.7	12	0.1813	0.1563	1.5104	0.0000	0.485	0.540
	0.8	12	0.2243	0.1800	1.4810	0.0000	0.553	0.545
	0.9	12	0.1276	0.1313	1.4929	0.2000	0.450	0.507
	target-specific	65	0.4503	0.3538	1.2087	0.4000	0.937	0.955
hif2a	generic	97,716	0.2577	0.2514	1.0236	0.2000	0.013	–
	0.1	95,840	0.3820	0.3053	0.9781	0.2000	0.021	0.783
	0.2	86,606	0.4367	0.3151	0.9647	0.2000	0.186	0.730
	0.3	65,880	0.2903	0.2097	1.0123	0.2000	0.016	0.322
	0.4	2,682	0.1134	0.0625	1.0951	0.0000	0.000	0.049
	0.5	87	0.2544	0.2394	1.0750	0.0000	0.054	0.477
	0.6	68	0.1482	0.2061	1.1039	0.2000	0.009	0.350
	0.7	64	-0.2187	-0.1451	1.3304	0.0000	0.001	0.001
	0.8	51	-0.1759	-0.1068	1.1562	0.0000	0.000	0.006
	target-specific	61	0.0296	0.0380	1.1956	0.0000	0.000	0.019
mapk14	generic	97,731	0.6817	0.4901	0.8706	0.4000	0.832	–
	0.1	83,734	0.6598	0.4614	0.8884	0.4000	0.731	0.280
	0.2	58,576	0.6548	0.4794	0.9567	0.2000	0.819	0.386
	0.3	23,722	0.6737	0.5835	0.9633	0.4000	0.980	0.952
	0.4	12,751	0.6628	0.5009	0.9584	0.0000	0.888	0.576
	0.5	11,345	0.6717	0.5009	0.8395	0.4000	0.871	0.601
	0.6	9,109	0.6771	0.5045	0.7786	0.4000	0.888	0.609
	0.7	1,950	0.5994	0.4650	0.8285	0.4000	0.770	0.373
	0.8	866	0.5550	0.3788	0.8410	0.4000	0.439	0.139
	0.9	600	0.4039	0.2926	0.9231	0.4000	0.204	0.057
	target-specific	1,713	0.3562	0.2065	1.1133	0.0000	0.070	0.005

Supplementary Table 5: ECIF::XGB results across proteins and training sets including: generic (0.0), target-specific, and family-specific thresholds. PCC, KT, RMSE, and Overlap@5 are taken from the selected-set metrics (no bootstrap resampling). Overlap@K is reported as Overlap@5. Both KT p-values are one-sided paired bootstrap comparisons (1000 resamples) on the same selected compounds: the first against the Generic AEV-PLIG reference model, the second against this same ECIF::RF family's own threshold-0.0 baseline which isolates target-specificity independent of architecture. $p < 0.05$ indicates the reference scored higher on Kτ in significantly more than half of paired resamples; $p > 0.95$ indicates this row's model scored significantly higher than the reference.

Dataset	MCL1	SYK	PFKFB3	HIF2α	MAPK14
0.9	119	754	12	3	600
0.8	122	7788	12	51	866
0.7	249	10978	12	64	1950
0.6	462	11432	12	68	9109
0.5	681	15231	12	87	11345
0.4	4167	18498	14	2682	12751
0.3	58702	34492	3097	65880	23722
0.2	82353	71837	31260	86606	58576
0.1	94478	88974	80921	95840	83734
0.0	97692	97686	97726	97716	97731

Supplementary Table 6: Database sizes of each Structural Similarity threshold with TM-align. Each database has been processed with RDKit, removing complexes that cannot be parsed due to uncommon elements or unknown bond types. Comparison of the original containing all protein complexes versus Target-specific Datasets for each protein of interest prior to 0.9 Tanimoto similarity filtering.

Protein	ΔG range	ΔG std	Mean Tanimoto	Cliff fraction	M'ful pockets	Mean drug*	Best drug*	Known inhibitors	Clinical Stage	Clinical Status
MCL1	4.180	1.066	0.404	0.000	1	0.908	0.908	1	Yes	Clinical trials
SYK	4.127	0.810	0.509	0.128	5	0.007	0.102	20	Yes	FDA approved
PFKFB3	3.717	1.089	0.529	0.244	3	0.062	0.868	5	Limited	Preclinical
HIF2A	4.560	1.071	0.372	0.200	1	0.865	0.865	1	Yes	FDA approved
MAPK14	3.780	1.008	0.499	0.261	3	0.268	0.750	1	Yes	Failed trials

Supplementary Table 7: Protein and binding-site characteristics for the five FEP benchmark targets. n, ΔG range, and ΔG std summarize the experimental congeneric series. Meaningful pockets, score concentration, best druggability, and best druggability pocket ID come from fpocket (old_scripts/protein_exploration/pocket.csv); Mean Tanimoto is the average pairwise Tanimoto similarity (2048-bit ECFP6) across the congeneric series; Cliff fraction is the fraction of structurally similar pairs (Tanimoto ≥ 0.7) that also differ in experimental ΔG by at least 1.364 kcal/mol ($\sim$1 $\log_{10}$ unit).

Threshold	Mean K_τ	Std	Min	Max	CI Low	CI High
0.0	0.458	0.054	0.327	0.548	0.416	0.558
0.1	0.455	0.058	0.341	0.566	0.416	0.547
0.2	0.462	0.082	0.145	0.576	0.406	0.571
0.3	0.433	0.074	0.215	0.538	0.352	0.503
0.4	0.234	0.118	-0.124	0.406	0.191	0.387
0.5	0.277	0.114	0.041	0.490	0.281	0.451
0.6	0.256	0.083	0.155	0.357	0.148	0.315
0.7	0.191	0.169	-0.343	0.429	0.157	0.335
0.8	0.187	0.148	-0.241	0.422	0.197	0.369
0.9	0.171	0.161	-0.438	0.448	0.084	0.269
target-specific	0.276	0.113	-0.048	0.441	0.279	0.445

Supplementary Table 8: AEV-PLIG::GATv2 target-matched threshold sweep and target-specific performance on MCL1. Mean, Std, Min, and Max are computed across the per-seed and per-fold. CI Low/High is a genuine bootstrap 95% confidence interval (1000 resamples, compound-level) computed by pooling the canonical prediction stream across every available fold for that row and bootstrapping once.